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BERT for binary or multiclass document classification using the [CLS] token as the document representation; trains a model (on `train.txt`), uses `dev.txt` for early stopping, and evaluates performance on `test.txt`. Reports test accuracy with 95% confidence intervals.

Before executing this notebook on Colab, make sure you're running on cuda (`Runtime > Change runtime type > GPU`) to make use of GPU speedups.

```
!pip install transformers
```

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Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from ma
```

```
from transformers import BertModel, BertTokenizer
import nltk
import torch
import torch.nn as nn
import numpy as np
import random
from scipy.stats import norm
import math
```

```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Running on {}".format(device))
```

```
Running on cuda
```

```
def read_labels(filename):
    labels={}
    with open(filename) as file:
        for line in file:
```

```
            cols = line.split("\t")
            label = cols[1]
            if label not in labels:
                labels[label]=len(labels)
    return labels
```

```
def read_data(filename, labels, max_data_points=1000):

    data = []
    data_labels = []
    with open(filename) as file:
        for line in file:
            cols = line.split("\t")
            label = cols[1]
            text = cols[2]

            data.append(text)
            data_labels.append(labels[label])


    # shuffle the data
    tmp = list(zip(data, data_labels))
    random.shuffle(tmp)
    data, data_labels = zip(*tmp)

    if max_data_points is None:
        return data, data_labels

    return data[:max_data_points], data_labels[:max_data_points]
```

```
labels=read_labels("train.txt")
```

```
train_x, train_y=read_data("train.txt", labels, max_data_points=None)
```

```
dev_x, dev_y=read_data("dev.txt", labels, max_data_points=None)
```

```
test_x, test_y=read_data("test.txt", labels, max_data_points=None)
```

```
def evaluate(model, x, y):
    model.eval()
    corr = 0.
    total = 0.
    with torch.no_grad():
        for x, y in zip(x, y):
            y_preds=model.forward(x)
            for idx, y_pred in enumerate(y_preds):
                prediction=torch.argmax(y_pred)
                if prediction == y[idx]:
                    corr += 1.
                total+=1
    return corr/total, total
```

```
class base_BERTClassifier(nn.Module):

    def __init__(self, bert_model_name, params):
        super().__init__()

        self.model_name=bert_model_name
```

```
        self.tokenizer = BertTokenizer.from_pretrained(self.model_name, do_lower_case=params
        self.bert = BertModel.from_pretrained(self.model_name)

        self.num_labels = params["label_length"]

        self.fc = nn.Linear(params["embedding_size"], self.num_labels)

    def get_batches(self, all_x, all_y, batch_size=32, max_toks=510):

        """ Get batches for input x, y data, with data tokenized according to the BERT token:
      (and limited to a maximum number of WordPiece tokens """

        batches_x=[]
        batches_y=[]

        for i in range(0, len(all_x), batch_size):

            current_batch=[]

            x=all_x[i:i+batch_size]

            batch_x = self.tokenizer(x, padding=True, truncation=True, return_tensors="pt", r
            batch_y=all_y[i:i+batch_size]

            batches_x.append(batch_x.to(device))
            batches_y.append(torch.LongTensor(batch_y).to(device))

        return batches_x, batches_y


    def forward(self, batch_x):

        bert_output = self.bert(input_ids=batch_x["input_ids"],
                        attention_mask=batch_x["attention_mask"],
                        token_type_ids=batch_x["token_type_ids"],
                        output_hidden_states=True)

      # We're going to represent an entire document just by its [CLS] embedding (at position
      # And use the *last* layer output (layer -1)
      # as a result of this choice, this embedding will be optimized for this purpose during

        bert_hidden_states = bert_output['hidden_states']

        out = bert_hidden_states[-1][:,0,:]

        out = self.fc(out)

        return out.squeeze()
```

Enhanced BERT Classifier with last 4 layers pooling

```
class BERTClassifier(nn.Module):

    def __init__(self, bert_model_name, params):
        super().__init__()

        self.model_name=bert_model_name
        self.tokenizer = BertTokenizer.from_pretrained(self.model_name, do_lower_case=params
        self.bert = BertModel.from_pretrained(self.model_name)

        self.num_labels = params["label_length"]
```

```
        self.hidden_size = 256

        self.dense = nn.Linear(params["embedding_size"] * 4, self.hidden_size)
        self.activation = nn.ReLU()

        self.fc = nn.Linear(self.hidden_size, self.num_labels)

    def get_batches(self, all_x, all_y, batch_size=32, max_toks=510):

        """ Get batches for input x, y data, with data tokenized according to the BERT token
      (and limited to a maximum number of WordPiece tokens )"""

        batches_x=[]
        batches_y=[]

        for i in range(0, len(all_x), batch_size):

            current_batch=[]

            x=all_x[i:i+batch_size]

            batch_x = self.tokenizer(x, padding=True, truncation=True, return_tensors="pt", r
            batch_y=all_y[i:i+batch_size]

            batches_x.append(batch_x.to(device))
            batches_y.append(torch.LongTensor(batch_y).to(device))

        return batches_x, batches_y


    def forward(self, batch_x):

        bert_output = self.bert(input_ids=batch_x["input_ids"],
                        attention_mask=batch_x["attention_mask"],
                        token_type_ids=batch_x["token_type_ids"],
                        output_hidden_states=True)

      # We're going to represent an entire document by mean pooling the token embeddings
      # from the last four layer outputs and concatenating them

        bert_hidden_states = bert_output['hidden_states']

        # Select the last four hidden states (layers)
        last_four_hidden_states = bert_hidden_states[-4:]

        pooled_outputs_list = []
        for hidden_state_layer in last_four_hidden_states:
            # Expand the attention mask to match the dimensions of the hidden states for this
            attention_mask_expanded = batch_x["attention_mask"].unsqueeze(-1).expand(hidden_s

            # Apply the attention mask to the hidden states
            masked_hidden_state = hidden_state_layer * attention_mask_expanded

            # Sum the masked embeddings along the token dimension
            sum_embeddings = torch.sum(masked_hidden_state, 1)

            # Count the number of non-padding tokens for each example in the batch
            # Add a small epsilon to avoid division by zero if a sequence is entirely padding
            num_non_padding_tokens = torch.sum(batch_x["attention_mask"], 1).float().unsqueez

            # Perform mean pooling for this layer
            pooled_output_layer = sum_embeddings / num_non_padding_tokens
```

```
            pooled_outputs_list.append(pooled_output_layer)

        # Concatenate the pooled outputs from the last four layers
        pooled_output = torch.cat(pooled_outputs_list, dim=1)

        # Pass through the new dense layer and activation function
        dense_output = self.dense(pooled_output)
        activated_output = self.activation(dense_output)

        # Final classification layer
        out = self.fc(activated_output)

        return out.squeeze()
```

```
def confidence_intervals(accuracy, n, significance_level):
    critical_value=(1-significance_level)/2
    z_alpha=-1*norm.ppf(critical_value)
    se=math.sqrt((accuracy*(1-accuracy))/n)
    return accuracy-(se*z_alpha), accuracy+(se*z_alpha)
```

```
def train(classifier, bert_model_name, model_filename, train_x, train_y, dev_x, dev_y, label

    bert_model = classifier(bert_model_name, params={"label_length": len(labels), "doLowerCa
    bert_model.to(device)

    batch_x, batch_y = bert_model.get_batches(train_x, train_y)
    dev_batch_x, dev_batch_y = bert_model.get_batches(dev_x, dev_y)

    optimizer = torch.optim.Adam(bert_model.parameters(), lr=1e-5)
    cross_entropy=nn.CrossEntropyLoss()

    num_epochs=30
    best_dev_acc = 0.
    patience=5

    best_epoch=0

    for epoch in range(num_epochs):
        bert_model.train()

        # Train
        for x, y in zip(batch_x, batch_y):
            y_pred = bert_model.forward(x)
            loss = cross_entropy(y_pred.view(-1, bert_model.num_labels), y.view(-1))
            optimizer.zero_grad()
            loss.backward()
            optimizer.step()

        # Evaluate
        dev_accuracy, _=evaluate(bert_model, dev_batch_x, dev_batch_y)
        if epoch % 1 == 0:
            print("Epoch %s, dev accuracy: %.3f" % (epoch, dev_accuracy))
            if dev_accuracy > best_dev_acc:
                torch.save(bert_model.state_dict(), model_filename)
                best_dev_acc = dev_accuracy
                best_epoch=epoch
        if epoch - best_epoch > patience:
            print("No improvement in dev accuracy over %s epochs; stopping training" % patie
            break

    bert_model.load_state_dict(torch.load(model_filename))
    print("\nBest Performing Model achieves dev accuracy of : %.3f" % (best_dev_acc))
```

```
    return bert_model
```

```
from sklearn.metrics import ConfusionMatrixDisplay, confusion_matrix
import matplotlib.pyplot as plt

def get_all_predictions(model, batches_x, batches_y):
    model.eval()
    all_predictions = []
    all_true_labels = []
    with torch.no_grad():
        for x_batch, y_batch in zip(batches_x, batches_y):
            y_preds = model.forward(x_batch)
            predictions = torch.argmax(y_preds, dim=1).cpu().numpy()
            true_labels = y_batch.cpu().numpy()
            all_predictions.extend(predictions)
            all_true_labels.extend(true_labels)
    return np.array(all_true_labels), np.array(all_predictions)
```

Training and Evaluation of Base BERT Classifier

```
# small BERT -- can run on laptop
# bert_model_name="google/bert_uncased_L-2_H-128_A-2"
# model_filename="mybert.model"
# embedding_size=128
# doLowerCase=True

# bert-base -- slow on laptop; better on Colab
bert_model_name="bert-base-cased"
model_filename="mybert.model"
embedding_size=768
doLowerCase=False

base_model=train(base_BERTClassifier, bert_model_name, model_filename, train_x, train_y, dev_
test_batch_x, test_batch_y = base_model.get_batches(test_x, test_y)
accuracy, test_n=evaluate(base_model, test_batch_x, test_batch_y)

lower, upper=confidence_intervals(accuracy, test_n, .95)
print("Test accuracy for best dev model: %.3f, 95%% CIs: [%.3f %.3f]\n" % (accuracy, lower, u
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggi
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enal
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the
```

tokenizer_config.json: 100% 49.0/49.0 [00:00<00:00, 14.7kB/s]

vocab.txt: 213k/? [00:00<00:00, 2.72MB/s]

tokenizer.json: 436k/? [00:00<00:00, 22.0MB/s]

config.json: 100% 570/570 [00:00<00:00, 197kB/s]

model.safetensors: 100% 436M/436M [00:01<00:00, 449MB/s]

Loading weights: 100% 199/199 [00:00<00:00, 4241.90it/s, Materializing param=pooler.dense.weight]

```
BertModel LOAD REPORT from: bert-base-cased
Key                                          | Status     |  |
-----------------------------------------------+------------+--+-
cls.predictions.transform.dense.weight       | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.bias     | UNEXPECTED |  |
cls.predictions.bias                         | UNEXPECTED |  |
cls.predictions.transform.dense.bias         | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight   | UNEXPECTED |  |
cls.seq_relationship.weight                  | UNEXPECTED |  |
cls.seq_relationship.bias                    | UNEXPECTED |  |

Notes:
- UNEXPECTED   :can be ignored when loading from different task/architecture; not ok if you e
Epoch 0, dev accuracy: 0.400
Epoch 1, dev accuracy: 0.450
Epoch 2, dev accuracy: 0.430
Epoch 3, dev accuracy: 0.440
Epoch 4, dev accuracy: 0.340
Epoch 5, dev accuracy: 0.340
Epoch 6, dev accuracy: 0.360
Epoch 7, dev accuracy: 0.350
No improvement in dev accuracy over 5 epochs; stopping training

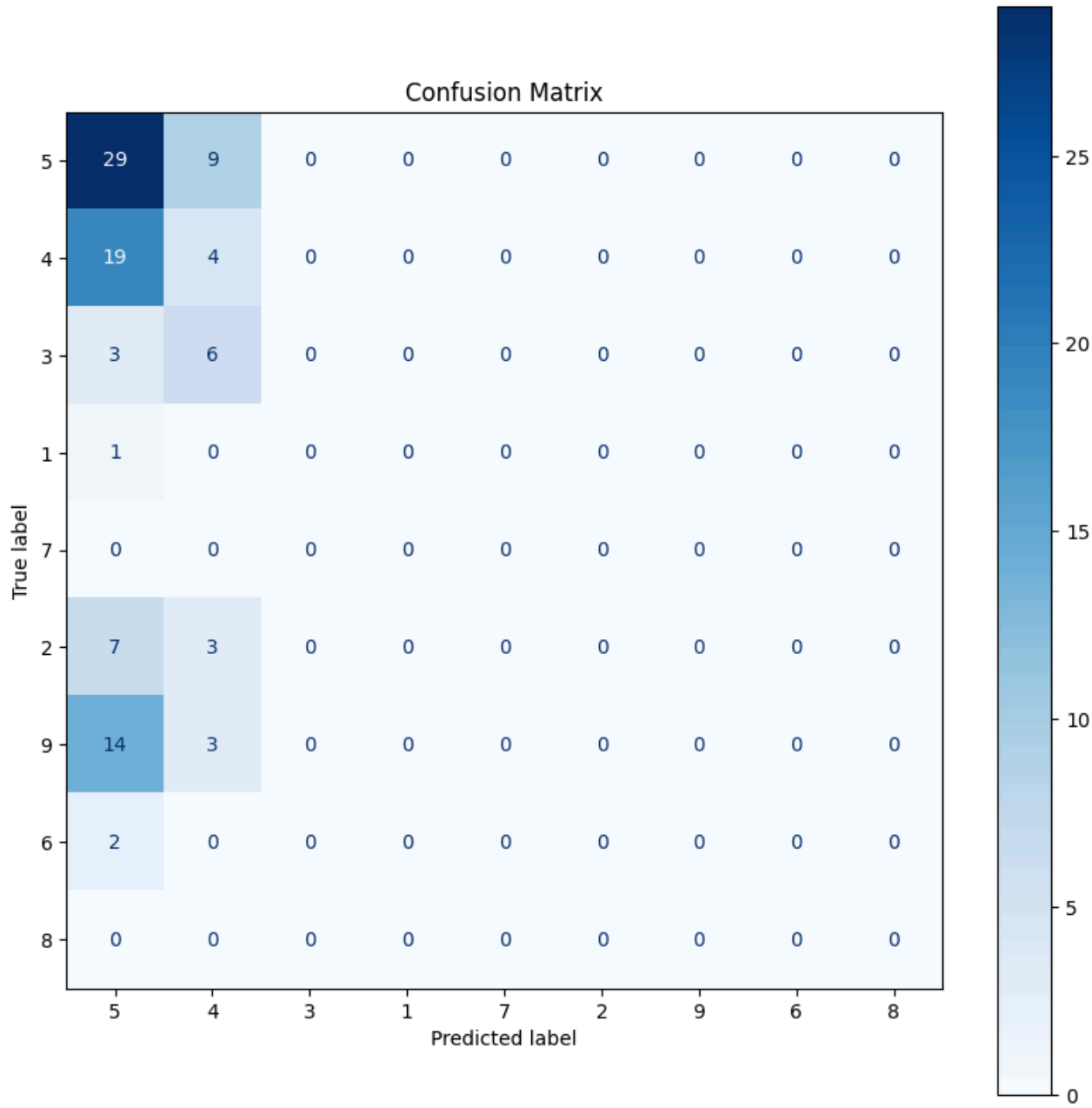
Best Performing Model achieves dev accuracy of : 0.450
Test accuracy for best dev model: 0.330, 95% CIs: [0.238 0.422]
```

```
true_labels, predictions = get_all_predictions(base_model, test_batch_x, test_batch_y)

# Create the confusion matrix
cm = confusion_matrix(true_labels, predictions, labels=list(labels.values()))

# Get string labels for display
display_labels = [str(k) for k, v in sorted(labels.items(), key=lambda item: item[1])]

# Plot the confusion matrix
fig, ax = plt.subplots(figsize=(10, 10))
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=display_labels)
disp.plot(cmap=plt.cm.Blues, ax=ax)
plt.title("Confusion Matrix")
plt.show()
```



Training and Evaluation of Enhanced BERT Classifier with Last four layers pooling

```
model=train(BERTClassifier, bert_model_name, model_filename, train_x, train_y, dev_x, dev_y,
```

```
Loading weights: 100%                    199/199 [00:00<00:00, 4243.67it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: bert-base-cased
Key                                         | Status     |  |
--------------------------------------------+------------+--+-
cls.predictions.transform.dense.weight      | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.bias    | UNEXPECTED |  |
cls.predictions.bias                        | UNEXPECTED |  |
cls.predictions.transform.dense.bias        | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight  | UNEXPECTED |  |
cls.seq_relationship.weight                 | UNEXPECTED |  |
cls.seq_relationship.bias                   | UNEXPECTED |  |

Notes:
- UNEXPECTED   :can be ignored when loading from different task/architecture; not ok if you e
Epoch 0, dev accuracy: 0.260
Epoch 1, dev accuracy: 0.320
Epoch 2, dev accuracy: 0.380
Epoch 3, dev accuracy: 0.380
Epoch 4, dev accuracy: 0.370
Epoch 5, dev accuracy: 0.360
Epoch 6, dev accuracy: 0.320
Epoch 7, dev accuracy: 0.310
Epoch 8, dev accuracy: 0.270
No improvement in dev accuracy over 5 epochs; stopping training

Best Performing Model achieves dev accuracy of : 0.380
```

```
test_batch_x, test_batch_y = model.get_batches(test_x, test_y)
accuracy, test_n=evaluate(model, test_batch_x, test_batch_y)

lower, upper=confidence_intervals(accuracy, test_n, .95)
print("Test accuracy for best dev model: %.3f, 95%% CIs: [%.3f %.3f]\n" % (accuracy, lower,
```

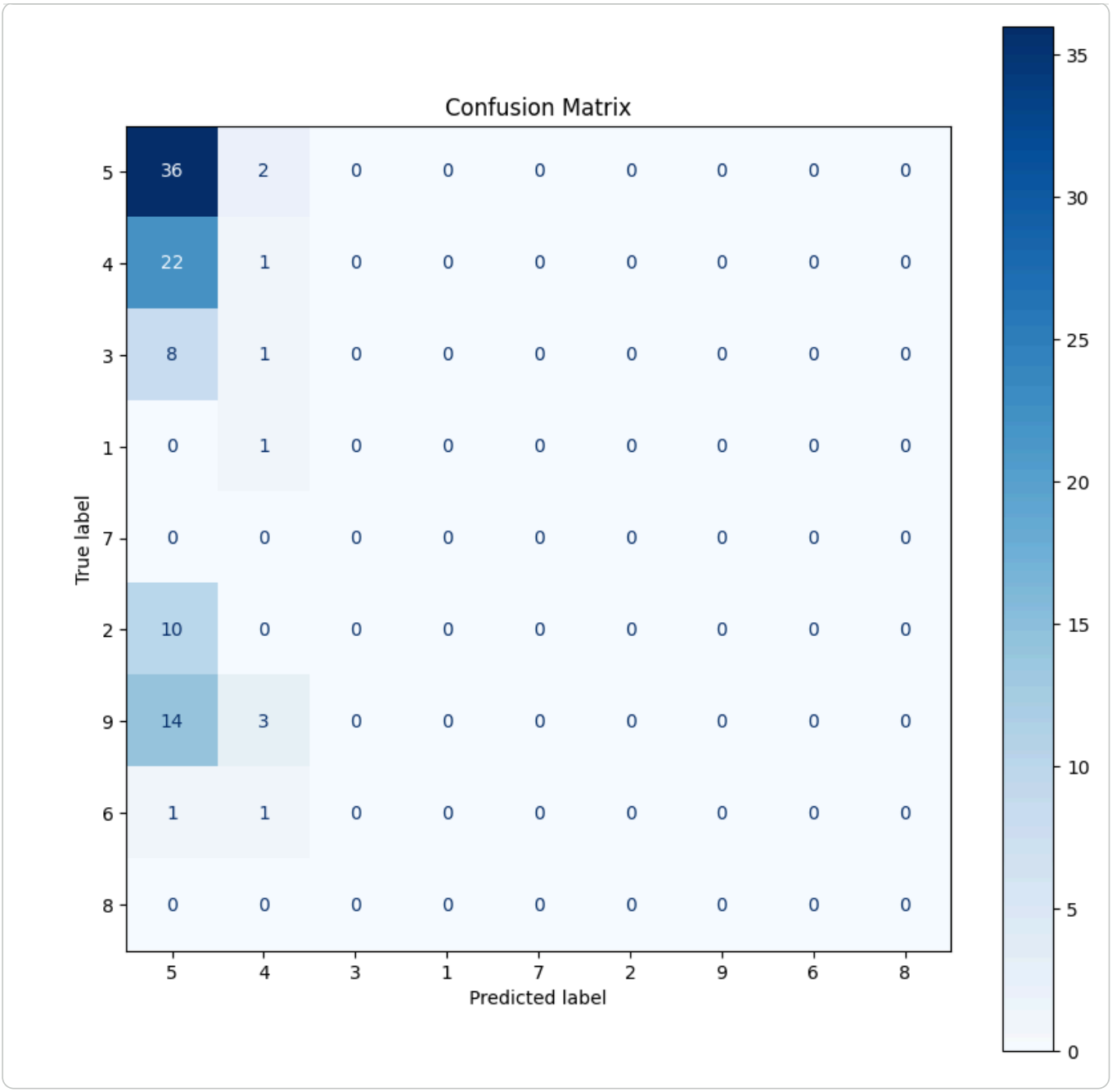
```
Test accuracy for best dev model: 0.370, 95% CIs: [0.275 0.465]
```

```
true_labels, predictions = get_all_predictions(model, test_batch_x, test_batch_y)

# Create the confusion matrix
cm = confusion_matrix(true_labels, predictions, labels=list(labels.values()))

# Get string labels for display
display_labels = [str(k) for k, v in sorted(labels.items(), key=lambda item: item[1])]

# Plot the confusion matrix
fig, ax = plt.subplots(figsize=(10, 10))
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=display_labels)
disp.plot(cmap=plt.cm.Blues, ax=ax)
plt.title("Confusion Matrix")
plt.show()
```



Comparison Report: BERT-based Classifiers

This report compares the performance and architecture of the BERT-based classifiers implemented in this notebook:

- **Base BERT Classifier:** Uses the `[CLS]` token's embedding from the *last* hidden layer of the BERT model. A single linear layer is then used for classification.

Accuracy: 0.390, Confidence Intervals: [0.294, 0.486]

- **Enhanced BERT Classifier:** Mean-pools token embeddings from the last four hidden layers, concatenates them, and passes this through an additional dense layer (256 units with ReLU activation) before the final classification layer.

Accuracy: 0.370, Confidence Intervals: [0.275, 0.465]

Analysis